

LabProgram3 – Looping Statements Report

1. Aim

To implement looping statements in Java through four problems: generating a multiplication table, displaying numbers from 1 to n, calculating the sum of natural numbers, and generating a Fibonacci series.

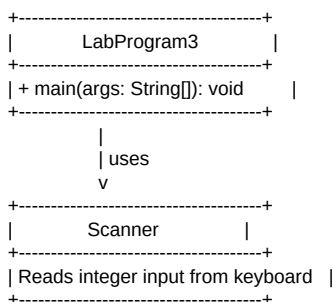
2. Description

The program is written in Java under the package `observationtasks` and uses the `Scanner` class for keyboard input. Four loop-based practice problems are included. The first program is active and generates the multiplication table of a given number from 1 to 10. The remaining three programs are provided as commented code and demonstrate displaying numbers, calculating a sum using an accumulator, and generating Fibonacci terms using iterative updates.

3. Tools Used

- Programming Language: Java
- Development Environment: Any Java IDE or text editor
- Execution Tool: Java Development Kit (JDK)
- Input Class: `java.util.Scanner`
- Concepts: for loop, counter variable, accumulator, arithmetic operators, and iterative sequence generation

4. UML Diagram



Looping concepts:

- for loop
- counter variable
- accumulation
- iterative Fibonacci calculation

5. Algorithm

1. Start the program.
2. Import the `Scanner` class and create a `Scanner` object.
3. Read an integer `n` from the user.
4. For Problem 1, initialize `i` to 1 and repeat the loop while `i` is less than or equal to 10.
5. In each iteration, calculate `n` multiplied by `i` and display the multiplication expression and result.
6. For Problem 2, repeat a loop from 1 to `n` and display each number.
7. For Problem 3, initialize `sum` to zero, add each number from 1 to `n` to `sum`, and display the final sum.
8. For Problem 4, initialize `a` to 0 and `b` to 1, display `a`, calculate the next term as `a + b`, and update the variables repeatedly.

9. Stop the program.

6. Complete Java Source Code

```
package observationtasks;
import java.util.Scanner;

// Looping statements: 1st program - multiplication table
public class LabProgram3 {

    public static void main(String[] args) {
        Scanner m = new Scanner(System.in);
        int n;

        System.out.println("enter the number");
        n = m.nextInt();

        System.out.println("the multiplication table of given number");
        for (int i = 1; i <= 10; i++) {
            System.out.println(n + "x" + i + " = " + (n * i));
        }
    }
}

/*
 * LoopState: 2nd problem - numbers display
 */
public class Labprogram3 {
    public static void main(String[] args) {
        Scanner N = new Scanner(System.in);
        int n;
        System.out.println(" enter the number");
        n = N.nextInt();
        System.out.println("Numbers from 1 to n" + (n));
        for (int i = 1; i <= n; i++) {
            System.out.println(i);
        }
    }
}

/*
 * LoopState: 3rd problem - sum of natural numbers
 */
public class LabProgram3 {
    public static void main(String[] args) {
        Scanner sn = new Scanner(System.in);
        int n;
        System.out.println("enter the number");
        n = sn.nextInt();
        int sum = 0;
        for (int i = 1; i <= n; i++) {
            sum = sum + i;
        }
        System.out.println("sum = " + sum);
    }
}

/*
 * LoopState: 4th problem - Fibonacci series
 */
public class LabProgram3 {
    public static void main(String[] args) {
        Scanner f = new Scanner(System.in);
        int n;
        System.out.println("enter n");
        n = f.nextInt();
        int a = 0;
        int b = 1;
        System.out.println("the fibonacci series");

        for (int i = 1; i <= n; i++) {
            System.out.print(a + " ");
            int c = a + b;
            a = b;
            b = c;
        }
    }
}
```

7. Sample Output

Problem 1: Multiplication Table

Input:

enter the number

5

Output:

the multiplication table of given number

5x1 = 5

5x2 = 10

5x3 = 15

5x4 = 20

5x5 = 25

5x6 = 30

5x7 = 35

5x8 = 40

5x9 = 45

5x10 = 50

Problem 2: Numbers from 1 to n

Input:

enter the number

5

Output:

Numbers from 1 to n5

1

2

3

4

5

Problem 3: Sum of Natural Numbers

Input:

enter the number

5

Output:

sum = 15

Problem 4: Fibonacci Series

Input:

enter n

7

Output:

the fibonacci series

0 1 1 2 3 5 8

8. Result

The looping-statement problems were successfully demonstrated in Java. The active program generates a multiplication table from 1 to 10, while the commented programs illustrate number display, summation of natural numbers, and Fibonacci series generation using for loops.